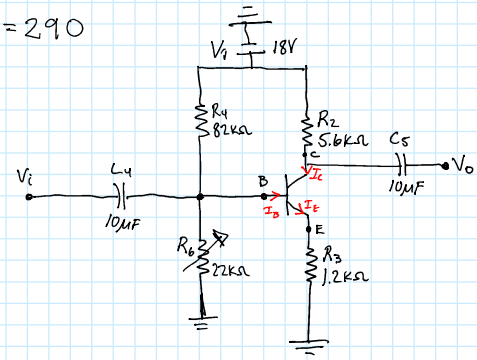
Lab 4

fredag 17. april 2026 13:20

Beregninger

$\beta = 290$



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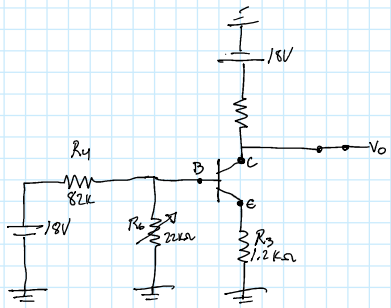

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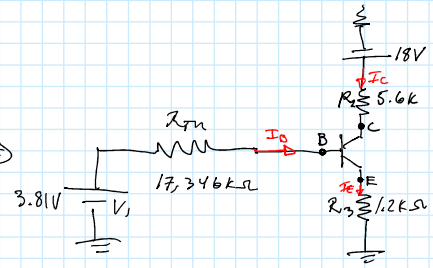


$$R_{Th} = \frac{R_4 \cdot R_6}{R_4 + R_6} = \frac{82k\Omega \cdot 22k\Omega}{82k\Omega + 22k\Omega}$$

$R_{Th} = 17{,}346\ k\Omega$

$$V_{Th} = V_1 \frac{R_6}{R_4 + R_6} = 18V \frac{22k\Omega}{82k\Omega + 22k\Omega}$$

$V_{Th} = 3{,}81V$

$$V_{Th} = R_{Th} I_B + V_{BE} + R_3 \cdot I_E$$

$$\Rightarrow I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (1+\beta)} = \frac{3{,}81V - 0.7V}{17346\Omega + 291}$$

A.

i. I følge input karakteristikken så ser man at en liten endring i V_{BE} gir en økning i I_B. Siden $I_C = \beta I_C$ gir dette en stor økning til I_C. → Når V_{BE} øker, øker I_C

ii.

$V_{CC} = R_C \cdot I_C + V_C$

$\Rightarrow V_C = V_{CC} - R_C \cdot I_C$

$V_C = 18V - 5.6k\Omega \cdot I_C$

Dersom I_C øker, så synker V_C

iii.

$V_E = R_E \cdot I_E$, $I_C \approx I_E$

$\Rightarrow V_E = R_E \cdot I_C$

Dersom I_C øker, øker V_E også

iv.

$V_{CC} = R_C I_C + V_{CE} + R_E I_E$, $I_E \approx I_C$

$\Rightarrow V_{CE} = V_{CC} - I_C(R_C + R_E)$

Dersom I_C øker, vil V_{CE} synke

B.

V_{BE} og I_C synker:

i. Når V_{BE} synker, så synker I_C ($I_C = \beta I_B$)

ii.

$I_C \downarrow \Rightarrow V_C \uparrow$

iii.

$I_C \downarrow \Rightarrow V_E \downarrow$

iv.

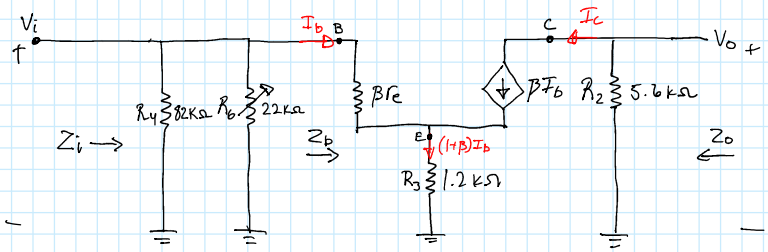
$I_C \downarrow \Rightarrow V_E \downarrow$

iv.

$I_C \downarrow \Rightarrow V_{CE} \uparrow$

C.

AC-ekvivalent krets



$$A_V = \frac{V_o}{V_i} = \frac{-Z_o \cdot I_c}{Z_b \cdot I_b} = \frac{-\beta Z_o \cancel{I_b}}{Z_b \cdot \cancel{I_b}} = \frac{-\beta Z_o}{Z_b}$$

$$Z_b = \frac{V_i}{I_b} = \frac{I_b \cdot \beta r_e + (1+\beta) I_b R_3}{I_b} \approx \frac{\beta \cancel{I_b}(r_e + R_3)}{\cancel{I_b}} = \beta(r_e + R_3)$$

$$Z_o = R_2$$

$$\Rightarrow A_V = -\frac{\cancel{\beta} R_2}{\cancel{\beta}(r_e + R_3)} = -\frac{R_2}{r_e + R_3}$$

$$R_3 \gg r_e \Rightarrow A_V = -\frac{R_2}{R_3} = -\frac{R_2}{1.2\,k\Omega}$$

R_2[kΩ]	A_V	Faseskift
4.7	3.92	180°
5.6	4.67	180°
6.8	5.67	180°

D.

180° faseskift mellom base og kollektor

E. C4:

Forhindre DC-spenning fra kilden, slik at det kun er AC-signalet som påvirker transistoren.

C5:

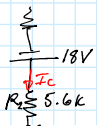
Gjør at kun AC sendes videre fra kretsen, og ikke DC-nivåer.

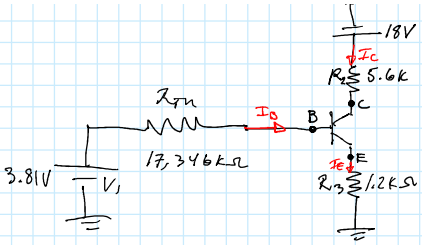
F.

Bypasser R_3:

$$Z_b = \frac{V_i}{I_b} = \frac{\cancel{I_b} \beta r_e}{\cancel{I_b}} = \beta r_e$$

$$A_V = \frac{V_o}{V_i} = \frac{-\beta I_b R_2}{Z_b \cdot I_b} = -\frac{\cancel{\beta} \cancel{I_b} R_2}{\cancel{\beta} r_e \cdot \cancel{I_b}} = -\frac{R_2}{r_e}$$





$$V_{Th} = R_{Th} I_B + V_{BE} + R_3 \cdot I_E$$

$$\Rightarrow I_B = \frac{V_{Th} - V_{BE}}{R_{Th} + (1+\beta) R_3} = \frac{3{,}81V - 0.7V}{17346\Omega + 291 \cdot 1.2k\Omega} = 8{,}48\,\mu A$$

$$I_E = (1+\beta) I_B = 291 \cdot 8{,}48\,\mu A = 2{,}47\,mA$$

$$\Rightarrow r_e = \frac{26mV}{I_E} = \frac{26mV}{2{,}47mA} = 10{,}53\,\Omega$$

$$A_V = -\frac{R_2}{10{,}53\Omega}$$

R_2[kΩ]	A_V	Faseskift
4.7	446	180°
5.6	531	180°
6.8	645	180°